Description

Background

Both in China and worldwide, the incidence of cancer is on the rise, posing a severe threat to human health and claiming lives at an alarming rate. Epidemiological data show that China's lifetime cancer incidence risk is about 27.6%, slightly higher than the global average of 25.1%—meaning roughly 1 in 4 people around the world face the threat of cancer in their lifetime. In terms of point prevalence rate, China reaches about 0.34% while the global rate is 0.67%, which fully highlights the urgent demand for effective cancer prevention, diagnosis and treatment strategies globally.

Among all malignant tumors, lung cancer remains the leading cause of cancer-related deaths, bringing heavy physical, psychological and economic burdens to patients, families and society. As a highly aggressive subtype of lung cancer, Small Cell Lung Cancer (SCLC) is a high-grade neuroendocrine carcinoma that has become one of the most intractable types of cancer to treat, mainly due to its high malignancy, rapid progression and tendency of early metastasis. Cigarette smoking is the primary risk factor for SCLC, with 95% of diagnosed patients having a history of tobacco use.

SCLC has distinct epidemiological and clinical characteristics that distinguish it from other lung cancer subtypes. In 2021, the incidence of SCLC in the United States was 4.7 cases per 100,000 individuals, and its 5-year overall survival rate is only 12% to 30%, far lower than that of non-small cell lung cancer. Clinically, SCLC is classified into limited stage (LS-SCLC, accounting for 30%) and extensive stage (ES-SCLC, accounting for 70%), depending on whether the disease can be covered by a radiation field usually confined to one hemithorax (which may include contralateral mediastinal and supraclavicular nodes). Notably, approximately 15% of patients have brain metastases at the time of diagnosis, further worsening the prognosis.

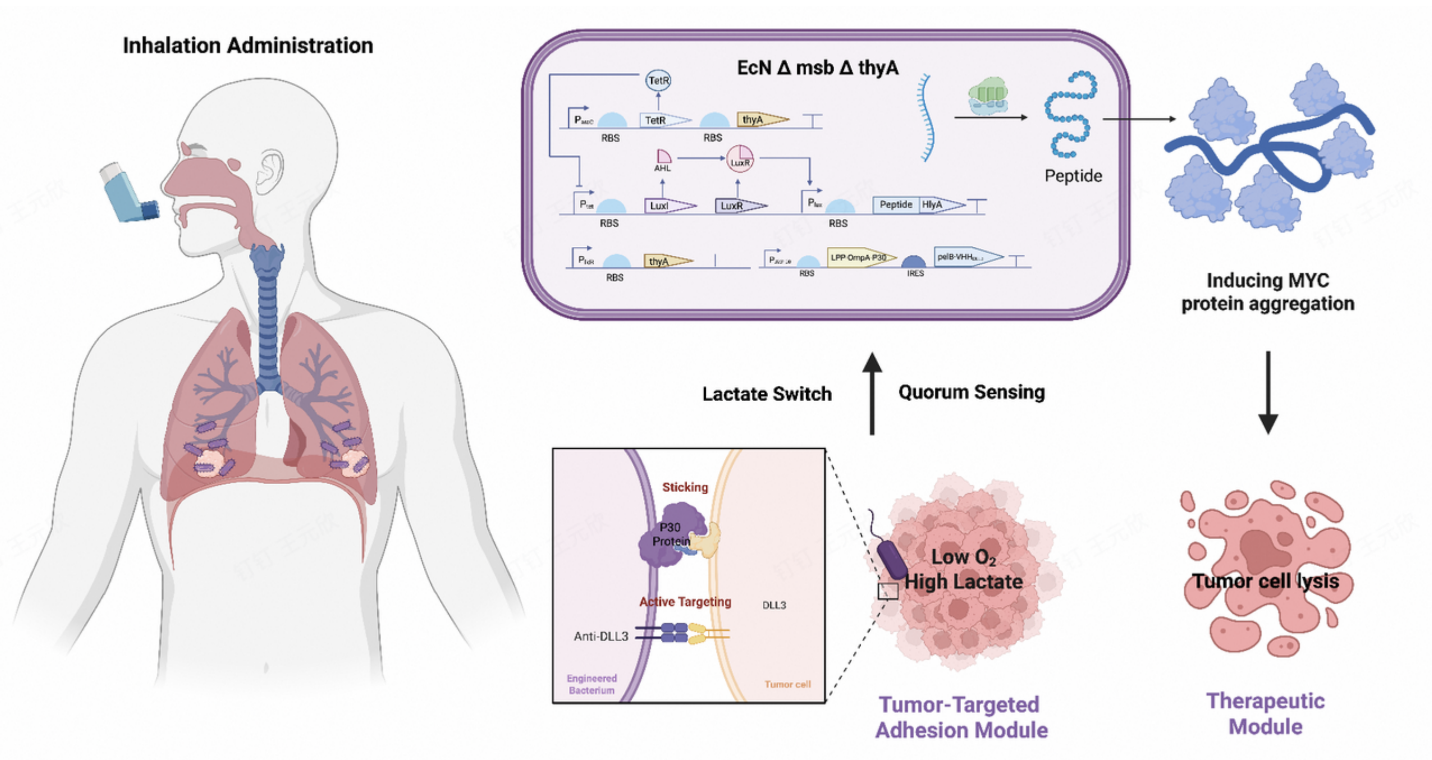
Despite decades of research, current treatments for SCLC still have obvious limitations and cannot meet clinical needs. The mainstream first-line treatment is a combination of platinum-etoposide chemotherapy and immunotherapy, which achieves an initial tumor shrinkage rate of 60% to 70%, but the therapeutic effect is unsustainable. Patients with ES-SCLC treated with this regimen have a median overall survival of only about 12 to 13 months, and 60% of them experience recurrence within 3 months, accompanied by prominent drug resistance problems.

Second-line therapies for ES-SCLC also have limited efficacy. Commonly used options include the DNA-alkylating agent lurbinectedin, which has an overall response rate of 35% and a median progression-free survival of 3.7 months, and tarlatamab, a bispecific T-cell engager targeting delta-like ligand 3, with an overall response rate of 40% and a median progression-free survival of 4.9 months—neither of which can significantly improve the long-term survival of patients. In addition, SCLC is difficult to screen early due to its aggressive clinical course and early metastasis; the limited use of clinical biomarkers, which is affected by genetic and epigenetic changes during treatment and the difficulty in obtaining high-quality tissue samples, further traps its treatment in a dilemma of "difficult diagnosis, difficult cure, high recurrence rate and high cost".

In summary, the increasing global cancer burden, the high mortality of lung cancer, and the unique aggressiveness and poor prognosis of SCLC, as well as the obvious shortcomings of current treatment methods in terms of efficacy and clinical application, indicate an urgent need for in-depth research on SCLC to develop more effective, safe and economical diagnosis and treatment strategies.

Project Overview

To address these unmet needs, our project devevelops an innovative way of killing cancer cells by using specially designed peptides to trigger protein aggregation in the cells. We takes *Escherichia coli Nissle 1917* (EcN) as the core chassis, leveraging its tumor-tropic and pulmonary compatibility advantages, and designs modular circuits including targeted delivery, pulmonary retention, precision targeting and stimuli-responsive therapy to achieve safe, specific and effective SCLC treatment while maintaining pulmonary homeostasis.



Chassis Selection: *Escherichia coli* Nissle 1917 (EcN)

Our project utilizes ***Escherichia coli* Nissle 1917 (EcN)** as the core chassis. As a clinically validated probiotic, EcN offers an exemplary safety profile and high human compatibility.

Crucially, EcN exhibits **innate tumor-tropic motility** , autonomously migrating toward the hypoxic and high-lactate tumor microenvironments (TME) characteristic of Small Cell Lung Cancer (SCLC). Since *E. coli* is a constituent of the native pulmonary microbiota, our engineered EcN is optimized to maintain **ecological homeostasis** while colonizing the respiratory tract.

Engineering Strategy and Modular Circuits

1. Targeted Delivery: Aerosolization & Particle Size Control

We employ **nebulization** as the primary administration route. By precisely calibrating the **aerodynamic diameter** of the aerosol droplets, we can physically control the deposition site of the engineered bacteria within the respiratory tree. This physical positioning ensures that the bacteria reach the deep bronchial or alveolar regions where SCLC resides, minimizing systemic exposure and enhancing local concentration.

2. Pulmonary Retention Module (Bronchial Adhesion)

Inspired by the specialized attachment mechanisms of *Mycoplasma pneumoniae* , we engineered EcN to heterologously express the **P30 adhesin** on its outer membrane. This module enables the bacteria to effectively bind to the bronchial epithelium, overcoming **mucociliary clearance** and ensuring stable, long-term colonization in the lung.

3. Homeostatic Regulation Module (Quorum Sensing & Negative Feedback)

To prevent bacterial overgrowth and potential immune overstimulation, we incorporated a **negative feedback loop** based on **Quorum Sensing (QS)** . When the bacterial density reaches a predefined threshold, the accumulated signaling molecules trigger a suppression circuit that slows down proliferation. This ensures a constant therapeutic dosage and maintains a stable population density within the lung.

4. Precision Targeting Module (DLL3-Modified OMVs)

The therapeutic engine relies on the secretion of **Outer Membrane Vesicles (OMVs)** . These vesicles are surface-modified with **DLL3 nanobodies** , which act as "homing devices" to recognize and bind to the Delta-like ligand 3 (DLL3) highly expressed on SCLC cells. This ensures that the therapeutic cargo is concentrated specifically at the tumor site.

5. Stimuli-Responsive Therapeutic Module (MMP-Triggered Peptides)

The OMVs encapsulate highly potent **cytotoxic short peptides** . To strictly prevent off-target toxicity, the expression of these peptides is regulated by a **Matrix Metalloproteinase (MMP) responsive promoter** . The drugs are synthesized and released only in the SCLC microenvironment, where MMP levels are significantly elevated, ensuring high spatial specificity.

6. Biosafety Module (Programmed Suicide Switch)

In compliance with iGEM's Gold Medal safety standards, we integrated a **programmed suicide switch** . Utilizing an inducible toxin-antitoxin system (eg, CcdB/CcdA), we can trigger total bacterial elimination upon completion of the treatment or in the event of accidental displacement, ensuring zero environmental leakage and long-term biosafety.

References

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